Haoying Zhang

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LinkedIn Profile | Github (Master projects) | GitLab (PhD projects)
Palaiseau, Ile de France - 91300, France

Education

- **Inria de Saclay and INSA** *September 2023 - August 2026*
  PhD in Computer Science Palaiseau, France
  - Title: Privacy-Preserving Analysis and Sharing of Time Series Data
  - **Supervisors:** Benjamin Nguyen, Nicolas Anciaux
- **Institut National des Sciences Appliquées (INSA)** *September 2018 - Juin 2023*
  Engineering Master's Degree (Double Degree with Master of Research in Cybersecurity) Rennes, France
  - Major: Computer Science
  - Grade: 15.17/20 (Rank: 3/74)

Experience

- **ENSTA Paris [🌐]** *September 2023 - Present*
  Teaching Assistant Palaiseau, France
  - Taught Oracle Database to second-year engineering students
  - Prepared materials (VMs, servers) and supported students with exercises and individual problem-solving
  - Co-supervised an master student's research internship at Inria
- **INSA CVL [🌐]** *Feburary 2023 - August 2023*
  Research Intern Bourges, France
  - Research internship with topic "Privacy-Preserving Graph Processing"
  - Developed a framework called PUZZLE to compute PageRank on large distributed graph in a privacy-preserving way (with Differential Privacy guarantee)
- **NAGRA NexGuard [🌐]** *May 2022 - August 2022*
  Intern Rennes, France
  - Engineering internship with topic "High Performance Computing for Forensic Watermarking Algorithms"
  - Implemented Single Instruction Multiple Data (SIMD) architectures with vector processors on x86 and ARM
  - Integrated SIMD architectures into forensic watermarking algorithms to achieve high performance on multiple platforms
- **Huawei [🌐]** *July 2021 - September 2021*
  Intern Beijing, China
  - Engineering internship with topic "Routing Protocol Testing"
  - Learnt 8+ TCP/IP routing protocols
  - Designed the topologies and implemented automatic tests

Publications

C=Conference, S=Submission

[S.1] Haoying Zhang, et al. (2026). **Privacy Assessment of Matrix Profiles Leak through Joint Reconstruction and Re-identification Attacks**. Under review at **NeurIPS** 2026.

[C.5] Charlène Jojon, Haoying Zhang, et al. (2026). **Verifiable Anomaly and Similarity Detection Using Matrix Profile in Private Time-series**. Accepted for publication in Australasian Conference on Information Security and Privacy **(ACISP)** 2026.

[C.4] Haoying Zhang, et al. (2026). **Privacy Attacks on Matrix Profiles via Reconstruction Techniques** [Artifact: Available, Functional, Reproduced]. Accepted for publication in **PETS** 2026.

[C.3] Haoying Zhang, et al. (2025). **Exploring Utility and Attackability Trade-offs in Local Differential Privacy**. Accepted for publication in **CCS** 2025.

[C.2] Haoying Zhang, et al. (2025). **TELESAFE: Monitoring Energy Consumption for Work-Life Boundaries in Telework**. Accepted for publication in **ICDM** 2025.

[C.1] Haoying Zhang, et al. (2025). **TELESAFE: Detecting Private/Work Boundary Crossings in Energy Consumption Trails in Telework**. Accepted for publication in **VLDB** 2025.

Skills

- **Programming Languages:** Assembly, C, C++, **Python**, Java, **Rust**, OCaml, Perl, Ruby, JavaScript, PHP
- **Database Systems:** MySQL, Oracle
- **Machine Learning:** Classification algorithms, RNNs, CNNs, GANs, etc
- **Cloud & Containerization:** Docker, Kubernetes
- **DevOps:** Git

Projects

- **Mycelium** *September 2021 - Juin 2022*
  Tools: Embedded system, C++, LoRaWAN, Raspberry Pi, Docker, Kubernetes [🌐]
  ◦ Implemented an alarm system in sensors to trigger actions based on weather scenarios
  ◦ Developed a Kubernetes-based fog infrastructure on a Raspberry Pi cluster for storing and processing sensor data transmitted via LoRaWAN, enabling monitoring of natural environments in Rennes
  ◦ Collaborated with scientists and engineers from Geosciences Rennes and Clermont-Ferrand University

- **Exchange Rate Reminder** *August 2020*
  Tools: Python, InfluxDB [GitHub]
  ◦ Developed a web-scraping application with data processing and storage in InfluxDB
  ◦ Implemented an email notification system triggered by exchange rate drops

- **Chessboard game** *August 2020*
  Tools: JavaFx [GitHub]

Volunteer Experience

- **Mathematics and Physics Tutor** *September 2018 - Feburary 2019*
  INSA
  ◦ Tutored fellow students in mathematics and physics through classes and Q&A sessions

- **Volunteer at the Planetarium** *August 2014*
  Beijing Planetarium
  ◦ Guided visitors through exhibits and provided telescope demonstrations

Additional Information

Languages: Chinese (Mother tongue), French (Fluent: DELF B2, 85/100), English (Fluent: TOEIC, 945/990)
Interests: Snowboarding, Badminton, Running, Astronomy